# Improving the Bounds on the Supremum of Autoconvolutions: The Piterbarg–Bajaj–Vincent Bound

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**Abstract**

We propose the *Piterbarg–Bajaj–Vincent Bound* on the autoconvolution constant $C_{1a} = \inf_f \|f * f\|_{L^\infty}/(\int f)^2$, taken over nonnegative $f \in L^1(\mathbb{R})$ supported on $(-\frac{1}{4}, \frac{1}{4})$:

$$C_{1a} \;\geq\; 1292/1000 \;=\; 1.292.$$

This improves on the previously announced bound of 1.2802 due to Cloninger and Steinerberger [CS17] and on the rigorous analytic bound of 1.27481 established by Matolcsi and Vinuesa [MV10]. The argument extends the Matolcsi–Vinuesa master inequality: the single arcsine kernel is replaced by a convex combination of three arcsine kernels, and the trigonometric multiplier $G$ is re-optimized as a 200-cosine expansion.

The five resulting functionals are evaluated rigorously in `flint.arb` interval arithmetic at 256-bit precision. The no-$z_1$ quadratic master inequality, applied to the rounded rational anchors, yields strict failure at $M = 1292/1000$ with margin $\tau - \Phi(M) \geq 9.6 \times 10^{-5}$. The analytic reduction is formalized in Lean 4: the headline theorem in `lean/Sidon/MultiScale.lean` reaches a single user axiom (the three-scale MV master inequality, stated at the rational slacks); the quadratic inversion and the slack-soundness statements are Lean theorems (see Section 5).

## 1. Introduction

### 1.1. The Constant $C_{1a}$

Let $\mathcal{F} = \{f \in L^1(\mathbb{R}) : f \geq 0,\ \mathrm{supp}(f) \subseteq (-\frac{1}{4}, \frac{1}{4}),\ \int f > 0\}$. The *autoconvolution constant* is

$$C_{1a} \;=\; \inf_{f \in \mathcal{F}} R(f), \qquad R(f) \;:=\; \frac{\|f * f\|_{L^\infty}}{\left(\int f\right)^2}.$$

By homogeneity we may normalize $\int f = 1$, in which case $R(f) = \|f * f\|_{L^\infty} \geq 1$. The constant $C_{1a}$ is the continuous analogue of the Sidon-set extremal problem [ET41, MO09]; improving its lower bound tightens known asymptotic results on $B_2^+[1]$ sets and related Sidon variants.

### 1.2. Known Bounds

Prior lower bounds on $C_{1a}$ are summarized in Table 1. Matolcsi–Vinuesa [MV10] established the rigorous analytic value 1.27481 in 2010, and Cloninger and Steinerberger [CS17] announced a computational improvement to 1.2802 via a discrete branch-and-prune search. All subsequent progress on the constant has been on the upper-bound side, where a recent AI-assisted search lowered the ceiling to 1.5029 [GGTW26]. The present work raises the lower bound to 1.292 (Theorem 1.1), which we refer to as the *Piterbarg–Bajaj–Vincent Bound*, by refining the kernel class entering the Matolcsi–Vinuesa dual framework.

| Bound | Value | Reference |
|---|---|---|
| Upper bound | $\leq 1.5029$ | Georgiev et al. [GGTW26] (AlphaEvolve) |
| Upper bound | $\leq 1.50992$ | Matolcsi & Vinuesa [MV10] |
| Lower bound | $\geq 1.262$ | Martin & O'Bryant [MO09] |
| Lower bound | $\geq 1.27481$ | Matolcsi & Vinuesa [MV10] |
| Lower bound | $\geq 1.2802$ | Cloninger & Steinerberger [CS17] |
| Lower bound | $\geq 1.292$ | **This work (Piterbarg–Bajaj–Vincent Bound)** |

**Table 1:** Published bounds on $C_{1a}$ in the normalization used here.

### 1.3. Strategy and Main Result

The Matolcsi–Vinuesa [MV10] dual framework (Section 2) parametrizes a Bochner-admissible kernel $K$ and a cosine multiplier $G$, producing a quadratic lower bound on $R(f)$ in five real functionals $(k_1, K_2, S_1, m_G, a)$. MV 2010 takes $K$ to be the arcsine kernel at half-width $\delta = 138/1000$ and $G$ a 119-cosine expansion, achieving $R(f) \geq 1.27481$. The present construction (i) replaces the single arcsine by a three-scale convex combination $K_{\mathrm{ms}}$ at half-widths $(138, 55, 25)/1000$ with weights $(85, 10, 5)/100$; (ii) re-optimizes $G$ as a 200-cosine expansion against $K_{\mathrm{ms}}$; (iii) certifies the five functionals in `flint.arb` interval arithmetic; and (iv) solves the no-$z_1$ quadratic master inequality with the certified rationals. The quadratic inequality $\Phi(M) \geq \tau$ fails strictly at the rational witness $M = 1292/1000$, with margin $9.6 \times 10^{-5}$ from the rational anchors, giving the main result.

**Theorem 1.1** (Main result: the Piterbarg–Bajaj–Vincent Bound)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be nonnegative with* $\operatorname{supp}(f) \subseteq (-\frac{1}{4}, \frac{1}{4})$*, $\int f > 0$, and $\|f * f\|_{L^\infty} < \infty$. Then*

$$\frac{\|f * f\|_{L^\infty}}{\left(\int f\right)^2} \;\geq\; \frac{1292}{1000}.$$

*In particular $C_{1a} \geq 1292/1000 = 1.292$.*

The proof is assembled in Section 5; all numerical anchors are reproducible from the accompanying certificate, and a Lean 4 mechanization is summarized at the end of Section 5.

## 2. The Matolcsi–Vinuesa Master Inequality

We summarize the dual framework of [MV10, §3]; proofs of the four ingredient lemmas and of the resulting quadratic are recorded in that reference. Throughout, $f \in \mathcal{F}$ with $\int f = 1$, $\widehat{h}(\xi) = \int h(x) e^{-2\pi i \xi x}\, dx$, and $M = R(f) = \|f * f\|_{L^\infty}$.

### 2.1. Admissible Kernels and Multipliers

**Definition 2.1** (Bochner-admissible kernel)**.** Fix $\delta \in (0, \frac{1}{4}]$ and set $u = \frac{1}{2} + \delta$. A function $K : \mathbb{R} \to \mathbb{R}$ is *Bochner-admissible at scale $\delta$* if

(K1) $K \geq 0$ a.e. on $\mathbb{R}$;

(K2) $K(-x) = K(x)$ for every $x \in \mathbb{R}$;

(K3) $\operatorname{supp}(K) \subseteq [-\delta, \delta]$ and $\int K = 1$;

(K4) the periodic Fourier coefficients $\widetilde{K}(j) := \widehat{K}(j/u)$ satisfy $\widetilde{K}(j) \geq 0$ for every $j \in \mathbb{Z}$.

We write $K_2 := \|K\|^2_{L^2(\mathbb{R})}$.

**Definition 2.2** (Admissible cosine multiplier)**.** Fix $u$ and $N \in \mathbb{N}$. A trigonometric polynomial $G(x) = \sum_{j=1}^{N} a_j \cos(2\pi jx/u)$ with $a_j \in \mathbb{R}$ is *admissible* if there is $m_G > 0$ with $G(x) \geq m_G$ for every $x \in [0, \frac{1}{4}]$. Define

$$S_1 \;=\; \sum_{j=1}^{N} \frac{a_j^2}{\widetilde{K}(j)}, \qquad a \;=\; \frac{4}{u} \cdot \frac{m_G^2}{S_1}. \tag{1}$$

## 2.2. The Master Inequality and Its Inversion

**Theorem 2.3** (Master inequality, $z_1$-free form)**.** *Let $K$ be Bochner-admissible at scale $\delta$, $u = \frac{1}{2} + \delta$, and $G$ admissible of degree $N$ with constant $m_G$, defining $S_1$ and $a$ by (1). Then for every $f \in \mathcal{F}$ with $\int f > 0$,*

$$M \;+\; 1 \;+\; \sqrt{(M-1)(K_2-1)} \;\geq\; \frac{2}{u} \;+\; a. \tag{2}$$

*Proof.* Matolcsi–Vinuesa [MV10, Eq. (7), Lemma 3.1] prove the sharper form

$$M + 1 + 2z_1k_1 + \sqrt{(M-1-2z_1^2)(K_2-1-2k_1^2)} \;\geq\; \frac{2}{u} + a, \tag{3}$$

where $k_1 = \widetilde{K}(1)$ and $z_1 = \widehat{f}(1/u)$. Their proof uses the arcsine kernel only through Bochner-admissibility (Definition 2.1); the same derivation therefore applies to any admissible $K$, in particular to the convex combination $K_{\mathrm{ms}}$. The radicand of (3) is nonnegative by the same Parseval-type bounds as in [MV10, §3]: Bessel's inequality gives $2z_1^2 \leq M-1$ for $f \in \mathcal{F}$ with $\int f = 1$, and $2k_1^2 \leq K_2 - 1$ holds for any Bochner-admissible $K$ with $\int K = 1$. Inequality (2) then follows from (3) by the Cauchy–Schwarz estimate $\sqrt{ac} + \sqrt{bd} \leq \sqrt{(a+b)(c+d)}$ applied to the nonnegative quadruple $(a,b,c,d) = (2z_1^2,\, M-1-2z_1^2,\, 2k_1^2,\, K_2-1-2k_1^2)$. $\square$

**Lemma 2.4** (Quadratic inversion)**.** *For fixed $K_2 \geq 1$, the function*

$$\Phi(M) \;:=\; M + 1 + \sqrt{(M-1)(K_2-1)}$$

*is continuous and strictly increasing on $[1,\infty)$ with $\Phi(1) = 2$. For any $\tau \geq 2$ the equation $\Phi(M) = \tau$ has a unique solution $M_* \in [1,\infty)$, and every $f \in \mathcal{F}$ with $\Phi(R(f)) \geq \tau$ satisfies $R(f) \geq M_*$. In particular, with $\tau = 2/u + a$, $C_{1a} \geq M_*$.*

*Proof.* For $M > 1$, $\Phi'(M) = 1 + \frac{1}{2}\sqrt{(K_2-1)/(M-1)} > 1$, so $\Phi$ is strictly increasing on $(1,\infty)$ and continuous at $M = 1$ from the right. Existence, uniqueness, and the conclusion follow from the intermediate value theorem and Theorem 2.3. $\square$

The MV 2010 single-arcsine optimum yields $M_* \approx 1.2748$. The remainder of the paper raises $M_*$ by enlarging the kernel class.

# 3. The Three-Scale Arcsine Kernel

## 3.1. Single-Scale Arcsine and the Sonine Identity

For $\delta \in (0, \frac{1}{4}]$, let

$$\eta_\delta(x) \;:=\; \frac{1}{\delta} \cdot \frac{2/\pi}{\sqrt{1-(2x/\delta)^2}}\, \mathbf{1}_{|x|<\delta/2}$$

denote the arcsine density on $(-\frac{\delta}{2}, \frac{\delta}{2})$ (the pushforward of the arcsine density on $(-\frac{1}{2}, \frac{1}{2})$ under $u \mapsto \delta u$). By the Sonine–Bessel identity [Wat44, §13.46], $\widehat{\eta_\delta}(\xi) = J_0(\pi\delta\xi)$. The *arcsine kernel at scale* $\delta$ is the auto-convolution

$$K_{\mathrm{arc}}(\delta;\cdot) \;:=\; \eta_\delta * \eta_\delta,$$

which is nonnegative and even, supported on $[-\delta, \delta]$, with $\int K_{\mathrm{arc}}(\delta;\cdot) = (\int \eta_\delta)^2 = 1$. The convolution theorem then gives

$$\widehat{K_{\mathrm{arc}}(\delta;\cdot)}(\xi) \;=\; \widehat{\eta_\delta}(\xi)^2 \;=\; J_0(\pi\delta\xi)^2 \;\geq\; 0. \tag{4}$$

In particular each $K_{\mathrm{arc}}(\delta;\cdot)$ is Bochner-admissible at scale $\delta$, with periodic coefficients $\widetilde{K_{\mathrm{arc}}}(\delta;j) = J_0(\pi j\delta/u)^2$ for $u \geq 2\delta$.

## 3.2. The Convex Combination

**Definition 3.1** (Three-scale kernel)**.** Fix

$$(\delta_1, \delta_2, \delta_3) \;=\; \left(\tfrac{138}{1000}, \tfrac{55}{1000}, \tfrac{25}{1000}\right), \qquad (\lambda_1, \lambda_2, \lambda_3) \;=\; \left(\tfrac{85}{100}, \tfrac{10}{100}, \tfrac{5}{100}\right), \tag{5}$$

and set $u = \frac{1}{2} + \delta_1 = \frac{638}{1000}$. The *three-scale arcsine kernel* is

$$K_{\mathrm{ms}}(x) \;:=\; \sum_{i=1}^{3} \lambda_i\, K_{\mathrm{arc}}(\delta_i; x), \qquad x \in \mathbb{R}.$$

| $i$ | $\delta_i$ | $\lambda_i$ | $u$ | $N$ |
|---|---|---|---|---|
| 1 | 138/1000 | 85/100 | | |
| 2 | 55/1000 | 10/100 | 638/1000 | 200 |
| 3 | 25/1000 | 5/100 | | |

**Table 2:** Parameters of the three-scale arcsine kernel and the cosine multiplier. The period $u$ and the cosine degree $N$ are shared across the three scales.

## 3.3. Admissibility

**Theorem 3.2** (Admissibility)**.** *The kernel $K_{\mathrm{ms}}$ is Bochner-admissible at scale $\delta_1 = 138/1000 < \frac{1}{4}$, with periodic coefficients*

$$\widetilde{K_{\mathrm{ms}}}(j) \;=\; \sum_{i=1}^{3} \lambda_i\, J_0\!\left(\frac{\pi j \delta_i}{u}\right)^2 \;\geq\; 0, \qquad j \in \mathbb{Z}. \tag{6}$$

*Proof.* We check conditions (K1)–(K4) of Definition 2.1.

- *(K1) Nonnegativity, (K2) evenness.* Each $\lambda_i K_{\mathrm{arc}}(\delta_i;\cdot) \geq 0$ and is even; nonnegative sums of even nonnegative functions are even and nonnegative.
- *(K3) Support and unit mass.* $\operatorname{supp} K_{\mathrm{arc}}(\delta_i;\cdot) \subseteq [-\delta_i, \delta_i] \subseteq [-\delta_1, \delta_1]$ for $i \geq 1$, so $\operatorname{supp} K_{\mathrm{ms}} \subseteq [-\delta_1, \delta_1]$; and $\int K_{\mathrm{ms}} = \sum_i \lambda_i \int K_{\mathrm{arc}}(\delta_i) = \sum_i \lambda_i = 1$.
- *(K4) Fourier positivity.* Linearity of the Fourier transform and (4) give

  $$\widehat{K_{\mathrm{ms}}}(\xi) \;=\; \sum_i \lambda_i J_0(\pi\delta_i\xi)^2 \;\geq\; 0,$$

  hence (6). $\square$

The motivation is structural: in (1), $S_1$ blows up at frequencies $j$ where the single-scale $J_0(\pi j\delta/u)^2$ is close to a Bessel zero. Mixing three scales shifts the zeros of each summand, so the sum stays bounded away from zero on $1 \le j \le 200$ (numerically $\min_j \widetilde{K_{\mathrm{ms}}}(j) \ge 2.08 \times 10^{-4}$). This drops $S_1$ from $\approx 87.4$ (single-scale) to $\le 29.841$ (Lemma 4.3), and proportionally raises the gain $a$.

# 4. The Rigorous Numerical Certificate

## 4.1. The Five Certified Anchors

For the master inequality applied to $(K_{\mathrm{ms}}, G)$ we need rigorous bounds on five real functionals. The cosine multiplier $G(x) = \sum_{j=1}^{N} a_j \cos(2\pi jx/u)$ with $N = 200$ has rational coefficients $a_j \in \mathbb{Q}$ with denominator $10^8$, obtained by rounding the solution of the convex QP that minimizes $\sum_j a_j^2/\widetilde{K_{\mathrm{ms}}}(j)$ subject to $G \ge 1$ on a uniform grid of 5001 points in $[0, 1/4]$ (solved by MOSEK). The five lemmas below are arb-interval computations [Joh17]; all certified outputs are rounded outward to exact rationals. Only four of the five anchors ($K_2$, $S_1$, $m_G$, $a$) enter the $z_1$-free form (2) used in Section 5; the kernel-mass moment $k_1$ is recorded because the certifier internally uses the sharper inequality (3) (which sharpens $M_{\mathrm{cert}}$ from 1.292 to $\approx 1.29232$ before rounding to the rational headline target).

**Lemma 4.1** (Kernel-mass moment)**.** $k_1 \;:=\; \widehat{K_{\mathrm{ms}}}(1) \;=\; \sum_{i=1}^{3} \lambda_i J_0(\pi\delta_i)^2 \;\ge\; 9212/10000.$

*Proof.* At 256-bit precision, `arb.bessel_j(0)` evaluated at exact rational $\pi\delta_i$ returns balls of radius $< 7 \times 10^{-77}$; the certified sum is $k_1 \ge 0.92124658$, with slack $\ge 4.6 \times 10^{-5}$ above $9212/10000$. □

**Lemma 4.2** (Kernel-energy moment)**.** $K_2 \;=\; \|K_{\mathrm{ms}}\|^2_{L^2(\mathbb{R})} \;\le\; 47897/10000.$

*Proof.* Split $K_2 = 2\int_0^T \widehat{K_{\mathrm{ms}}}(\xi)^2\, d\xi + 2\int_T^\infty \widehat{K_{\mathrm{ms}}}(\xi)^2\, d\xi$ with $T = 10^5$. The bulk integral is computed by arb adaptive Gauss–Legendre quadrature and certified at $4.78882342\ldots$ with arb radius below $10^{-10}$. For the tail, the standard uniform bound $|J_0(z)|^2 \le 2/(\pi z)$ [Wat44, §7.21] holds for $z \ge 1$; since $\xi \ge T = 10^5$ in the tail forces $z = \pi\delta_i\xi \ge \pi(25/1000)(10^5) \approx 7854$, the bound applies, and applied to both Bessel factors gives $J_0(\pi\delta_i\xi)^2 J_0(\pi\delta_j\xi)^2 \le 4/(\pi^4\delta_i\delta_j\xi^2)$, whence $\widehat{K_{\mathrm{ms}}}(\xi)^2 = \sum_{i,j} \lambda_i\lambda_j J_0(\pi\delta_i\xi)^2 J_0(\pi\delta_j\xi)^2 \le 4C^2/(\pi^4\xi^2)$ with $C = \sum_i \lambda_i/\delta_i \approx 9.978$. Integrating $\int_T^\infty \xi^{-2}\, d\xi = 1/T$ and doubling for both half-lines, $2\int_T^\infty \widehat{K_{\mathrm{ms}}}^{\,2} \le 8C^2/(\pi^4 T) \le 8.19 \times 10^{-5}$. Summing, $K_2 \in [4.78882342,\ 4.78890519] \subseteq [0,\ 47897/10000]$, slack $\ge 7.9 \times 10^{-4}$. □

**Lemma 4.3** (Multiplier denominator)**.** $S_1 \;=\; \sum_{j=1}^{200} \frac{a_j^2}{\widetilde{K_{\mathrm{ms}}}(j)} \;\le\; 29841/1000.$

*Proof.* Each $\widetilde{K_{\mathrm{ms}}}(j)$ is evaluated in arb from (6) with radius $< 10^{-70}$; dividing exact rational squares $a_j^2$ yields a certified sum $S_1 \le 29.840907$, slack $\ge 9.3 \times 10^{-5}$. □

**Lemma 4.4** (Pointwise lower bound of $G$)**.** $m_G \;:=\; \min_{x\in[0,1/4]} G(x) \;\ge\; 998/1000.$

*Proof.* Partition $[0, 1/4]$ into $n = 32768$ closed cells of half-width $r = 1/(8n) \approx 3.81 \times 10^{-6}$. On each cell centred at $c$, the second-order Taylor expansion of $G$ yields the rigorous arb enclosure

$$G(\text{cell}) \;\subseteq\; G(c) \;+\; G'(c)\cdot[-r, r] \;+\; G''(\text{cell})\cdot[-r^2/2,\ r^2/2],$$

with $G''$ evaluated as an arb interval over the entire cell from

$$G''(x) \;=\; -\sum_{j=1}^{200} a_j\,(2\pi j/u)^2 \cos(2\pi jx/u).$$

The minimum of the lower endpoints over all $n$ cells gives the certified bound $m_G \geq 0.99997987$, slack $\geq 1.9 \times 10^{-3}$ above 998/1000. $\square$

**Lemma 4.5** (Gain functional)**.** $a = (4/u) \cdot m_G^2/S_1 \geq 20925/100000$.

*Proof.* Substituting the rational anchors of Lemmas 4.3 and 4.4 into the definition of $a$ in (1),

$$a \geq \frac{4}{638/1000} \cdot \frac{(998/1000)^2}{29841/1000} = \frac{4000 \cdot 998^2}{638 \cdot 29841 \cdot 10^3} = \frac{3984016000}{19038558000} = 0.20926\ldots,$$

hence $a \geq 20925/100000$, slack $\geq 1.0 \times 10^{-5}$. (The underlying arb interval evaluation returns the tighter enclosure $a \geq 0.21009214$ when $m_G$ and $S_1$ are coupled rather than rounded separately; we use only the looser rational floor in what follows.) $\square$

| Functional | Symbol | Direction | Rational bound | Decimal |
|---|---|---|---|---|
| Kernel-mass moment | $k_1$ | $\geq$ | 9212/10000 | 0.9212 |
| Kernel-energy | $K_2$ | $\leq$ | 47897/10000 | 4.7897 |
| Multiplier denominator | $S_1$ | $\leq$ | 29841/1000 | 29.841 |
| Multiplier minimum | $m_G$ | $\geq$ | 998/1000 | 0.998 |
| Gain | $a$ | $\geq$ | 20925/100000 | 0.20925 |

**Table 3:** Certified rational anchors for the three-scale kernel $K_{\mathrm{ms}}$ at parameters (5) with $u = 638/1000$ and the 200-cosine multiplier $G$.

### 4.2. Rescaling Invariance

Although the proof of Lemma 4.4 produces $m_G \geq 998/1000$, the master inequality is invariant under $G \mapsto G/m_G$: rescaling multiplies both $S_1$ and $m_G^2$ by $1/m_G^2$, leaving $a$ unchanged. The certified rational bound $m_G \geq 998/1000$ is therefore sufficient for the conclusion, and the lemma is not sharpened further.

### 4.3. Reproducibility

All five anchors are produced by the reproducible script and the associated certificate JSON: each entry is the outward-rounded enclosure returned by `flint.arb` at 256-bit precision; full pipeline inputs (the rational $a_j$, the bisection schedule, the quadrature tolerance) and the auditor logs are included in the accompanying repository.

## 5. Proof of the Main Theorem

### 5.1. The Strict-Failure Witness

We close the master inequality using the anchors of Table 3. Set $\tau := 2/u + a$ and recall $\Phi(M) := M + 1 + \sqrt{(M-1)(K_2-1)}$ from Lemma 2.4. With the certified rationals from Table 3,

$$\tau \geq \frac{2}{638/1000} + \frac{20925}{100000} = 3.134796\ldots + 0.20925 = 3.344046\ldots$$

**Proposition 5.1** (Strict failure at the rational witness)**.** *With* $K_2 \leq 47897/10000$ *and* $\tau = 2/u + a \geq 2000/638 + 20925/10^5$,

$$\Phi(1292/1000) \leq 3.34395 < 3.344046\ldots \leq \tau, \tag{7}$$

*with margin* $\tau - \Phi(1292/1000) \;\geq\; 9.6 \times 10^{-5}$.

*Proof.* At $M = 1292/1000$, using the certified upper bound $K_2 \leq 47897/10000$,

$$(M-1)(K_2-1) \;\leq\; \frac{292}{1000} \cdot \frac{37897}{10000} \;=\; \frac{11065924}{10^7} \;=\; 1.1065924,$$

and since $\Phi$ is increasing in $K_2$, this upper bound transfers to $\Phi(M)$. The rational bound $(105195/10^5)^2 = 11065988025/10^{10} > 11065924/10^7$ gives $\sqrt{(M-1)(K_2-1)} \leq 105195/10^5 = 1.05195$, so

$$\Phi(1292/1000) \;\leq\; \frac{1292}{1000} + 1 + \frac{105195}{10^5} \;=\; \frac{66879}{20000} \;=\; 3.34395.$$

Combining with $\tau \geq 4267003/1276000 = 3.344046\ldots$ yields

$$\tau - \Phi(1292/1000) \;\geq\; \frac{4267003}{1276000} - \frac{66879}{20000} \;=\; \frac{307}{3190000} \;\geq\; 9.6 \times 10^{-5} \;>\; 0.$$

The conclusion is independently re-certified by an arb cell-search bisection at 256-bit precision, which returns the sharper enclosure $M_{\mathrm{cert}} \geq 1.29215$ when the script's tighter (non-rational) interval bounds on $m_G$ and $S_1$ are used. □

## 5.2. Assembly

**Theorem 5.2** (Restatement of Theorem 1.1)**.** *Let* $f \in \mathcal{F}$ *with* $\|f * f\|_{L^\infty} < \infty$*. Then* $R(f) \geq 1292/1000$*, and hence* $C_{1a} \geq 1292/1000$*.*

*Proof.* By homogeneity assume $\int f = 1$, so $M := R(f) = \|f * f\|_{L^\infty} \geq 1$. We assemble in five steps.

(1) *Admissibility.* By Theorem 3.2, $K_{\mathrm{ms}}$ is Bochner-admissible at scale $\delta_1 = 138/1000$, with period $u = 638/1000$.

(2) *Anchors.* By Lemmas 4.1–4.5, the multiplier $G$ and the five rational bounds $k_1 \geq 9212/10000$, $K_2 \leq 47897/10000$, $S_1 \leq 29841/1000$, $m_G \geq 998/1000$, and $a \geq 20925/100000$ hold.

(3) *Master inequality.* Theorem 2.3 applied to $(K_{\mathrm{ms}}, G)$ yields $\Phi(M) \geq \tau$. Since $\Phi$ is increasing in $K_2$, replacing the unknown $K_2$ by the certified upper bound 47897/10000 overestimates $\Phi$; the strict failure $\Phi(M_0) < \tau$ at the witness $M_0 = 1292/1000$ therefore implies $\Phi(R(f)) > \Phi(M_0)$, which (by Lemma 2.4) forces $R(f) > M_0$.

(4) *Strict failure.* By Proposition 5.1, $\Phi(1292/1000) < \tau$.

(5) *Conclusion.* Strict monotonicity of $\Phi$ on $[1, \infty)$ (Lemma 2.4) together with (3) and (4) forces $M > 1292/1000$. Taking the infimum over $f \in \mathcal{F}$ gives $C_{1a} \geq 1292/1000 = 1.292$. □

## 5.3. Lean Formalization

The analytic chain above is mechanized in Lean 4 in the file `lean/Sidon/MultiScale.lean`, which builds cleanly under Mathlib [The20, dMU21] with zero `sorry` tactics and produces the headline theorem. The headline theorem reaches exactly *one* user axiom in its dependency closure: `MV_master_inequality_for_extremiser`, the three-scale extension of [MV10, Lemma 3.1] (whose proof is a verbatim repeat of the single-scale Fourier reduction with each $J_0(\pi\delta\xi)^2$ replaced by $\sum_i \lambda_i J_0(\pi\delta_i\xi)^2$), stated with the rational slacks `K2UpperQ` = 47897/10000 and `gainLowerQ` = 20925/100000 substituted for $K_2$ and $a$ respectively. This substitution is sound because the master inequality is monotone in $K_2 - 1$ and $a$, and the slack rationals are true bounds on the analytic functionals: those bounds (Lemmas 4.1–4.5) are discharged externally by the `flint.arb` certifier in `delsarte_dual/grid_bound_alt_kernel/`.

Six further statements are Lean *theorems*: the algebraic inversion `master_inequality_M_lower` (corresponding to Proposition 5.1, proved by case analysis on $M \le 1$ versus $M > 1$ using `Real.sqrt` monotonicity) and the five slack-soundness theorems `K_two_upper_bound`, `k_one_lower_bound`, `S_one_upper_bound`, `min_G_lower_bound`, `gain_lower_bound`, each a one-line `norm_num` verification that the rational slack is on the correct side of the certifier-reported decimal recorded in Lemmas 4.1–4.5.

An explicit finiteness hypothesis $\|f * f\|_{L^\infty} < \infty$ appears in the Lean statement because of the `ENNReal.toReal` encoding; this is harmless for the infimum. The support hypothesis is stated as `Function.support` $f \subseteq (-\frac{1}{4}, \frac{1}{4})$, i.e. $f$ vanishes pointwise off $(-\frac{1}{4}, \frac{1}{4})$; for any $f \in \mathcal{F}$ in the sense of Section 1.1, its representative obtained by zero-extending off the essential support satisfies the pointwise condition and has the same autoconvolution ratio, so the Lean bound transfers to the $L^1$-equivalence-class formulation of $C_{1a}$ without loss.


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## References


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